$C_{\phi u}^{(12)}$ i1_i2_ $\rightarrow$

$$
\begin{aligned}
&\frac{1}{16\pi^2}\Big(\frac{2}{9}\tilde{\mu}\, g_1^2\, \overline{y_d}^{pr}\, a_d^{pr}\, \mathrm{LF}_{2,2,0}\left[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}\right]\delta_{i1i2} - \frac{1}{9}\tilde{\mu}\, g_1^2\, \overline{y_d}^{pr}\, a_d^{pr}\, \mathrm{LF}_{3,2,-1}\left[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}\right]\delta_{i1i2} + \\
&\frac{2}{9}\tilde{\mu}\, g_1^2\, \overline{y_e}^{pr}\, a_e^{pr}\, \mathrm{LF}_{2,2,0}\left[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}\right]\delta_{i1i2} - \frac{1}{9}\tilde{\mu}\, g_1^2\, \overline{y_e}^{pr}\, a_e^{pr}\, \mathrm{LF}_{3,2,-1}\left[m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}\right]\delta_{i1i2} - \\
&\frac{2}{9}\tilde{\mu}\, g_1^2\, \overline{y_e}^{pr}\, a_e^{pr}\, \mathrm{LF}_{3,1,0}\left[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}\right]\delta_{i1i2} - \frac{2}{9}\tilde{\mu}\, g_1^2\, \overline{y_e}^{pr}\, a_e^{pr}\, \mathrm{LF}_{3,2,-1}\left[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}\right]\delta_{i1i2} + \\
&\frac{1}{2}\tilde{\mu}\, g_1^2\, \overline{y_e}^{pr}\, a_e^{pr}\, \mathrm{LF}_{4,1,-1}\left[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}\right]\delta_{i1i2} - \frac{2}{9}\tilde{\mu}\, g_1^2\, \overline{y_e}^{pr}\, a_e^{pr}\, \mathrm{LF}_{5,1,-2}\left[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}\right]\delta_{i1i2} - \\
&\frac{2}{9}\tilde{\mu}\, g_1^2\, \overline{y_d}^{pr}\, a_d^{pr}\, \mathrm{LF}_{3,1,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}\right]\delta_{i1i2} - \frac{2}{9}\tilde{\mu}\, g_1^2\, \overline{y_d}^{pr}\, a_d^{pr}\, \mathrm{LF}_{3,2,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}\right]\delta_{i1i2} + \\
&\frac{5}{6}\tilde{\mu}\, g_1^2\, \overline{y_d}^{pr}\, a_d^{pr}\, \mathrm{LF}_{4,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}\right]\delta_{i1i2} - \frac{2}{3}\tilde{\mu}\, g_1^2\, \overline{y_d}^{pr}\, a_d^{pr}\, \mathrm{LF}_{5,1,-2}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}\right]\delta_{i1i2} - \\
&\frac{1}{9}\tilde{\mu}\, g_1^2\, \overline{y_u}^{pr}\, a_u^{pr}\, \mathrm{LF}_{2,2,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right]\delta_{i1i2} + \frac{1}{18}\tilde{\mu}\, g_1^2\, \overline{y_u}^{pr}\, a_u^{pr}\, \mathrm{LF}_{3,2,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right]\delta_{i1i2} + \\
&\frac{1}{9}\tilde{\mu}\, g_1^2\, \overline{y_u}^{pr}\, a_u^{pr}\, \mathrm{LF}_{3,1,0}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\right]\delta_{i1i2} + \frac{1}{9}\tilde{\mu}\, g_1^2\, \overline{y_u}^{pr}\, a_u^{pr}\, \mathrm{LF}_{3,2,-1}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\right]\delta_{i1i2} + \\
&\frac{1}{3}\tilde{\mu}\, g_1^2\, \overline{y_u}^{pr}\, a_u^{pr}\, \mathrm{LF}_{4,1,-1}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\right]\delta_{i1i2} - \frac{2}{3}\tilde{\mu}\, g_1^2\, \overline{y_u}^{pr}\, a_u^{pr}\, \mathrm{LF}_{5,1,-2}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\right]\delta_{i1i2} + \\
&\frac{1}{6} m_1\,\tilde{\mu}\, g_1^4\, \mathrm{LF}_{4,1,-1}[\tilde{\mu}, m_1]\,\delta_{i1i2} - \frac{2}{9} m_1\,\tilde{\mu}\, g_1^4\, \mathrm{LF}_{5,1,-2}[\tilde{\mu}, m_1]\,\delta_{i1i2} + \\
&\frac{1}{2} m_2\,\tilde{\mu}\, g_1^2\, g_2^2\, \mathrm{LF}_{4,1,-1}[\tilde{\mu}, m_2]\,\delta_{i1i2} - \frac{2}{3} m_2\,\tilde{\mu}\, g_1^2\, g_2^2\, \mathrm{LF}_{5,1,-2}[\tilde{\mu}, m_2]\,\delta_{i1i2} - \\
&\frac{1}{18} m_1\,\tilde{\mu}\, g_1^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, \mathrm{LF}_{2,1,1,0}\left[m_1, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{i1}\right] - \frac{1}{36} m_1\,\tilde{\mu}\, g_1^2\, \overline{y_u}^{pi1}\, y_u^{pi2} \\
&\quad \mathrm{LF}_{2,2,1,-1}\left[m_1, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{i1}\right] + \frac{1}{18} m_1\,\tilde{\mu}\, g_1^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, \mathrm{LF}_{3,1,1,-1}\left[m_1, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{i1}\right] - \\
&\frac{1}{12} m_1\,\tilde{\mu}\, g_1^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, \mathrm{LF}_{2,1,1,0}\left[m_1, m_{\tilde{q}}^{p}, \tilde{\mu}\right] + \frac{1}{12} m_1\,\tilde{\mu}\, g_1^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, \mathrm{LF}_{3,1,1,-1}\left[m_1, m_{\tilde{q}}^{p}, \tilde{\mu}\right] + \\
&\frac{1}{36} m_1\,\tilde{\mu}\, g_1^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, \mathrm{LF}_{2,2,1,-1}\left[m_1, m_{\tilde{u}}^{i1}, m_{\tilde{q}}^{p}\right] - \frac{1}{12} m_1\,\tilde{\mu}\, g_1^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, \mathrm{LF}_{2,2,1,-1}\left[m_1, \tilde{\mu}, m_{\tilde{q}}^{p}\right] - \\
&\frac{1}{6} m_1\,\tilde{\mu}\, g_1^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, \mathrm{LF}_{2,2,1,-1}\left[m_1, \tilde{\mu}, m_{\tilde{u}}^{i1}\right] + \frac{3}{4} m_2\,\tilde{\mu}\, g_2^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, \mathrm{LF}_{2,1,1,0}\left[m_2, m_{\tilde{q}}^{p}, \tilde{\mu}\right] - \\
&\frac{3}{4} m_2\,\tilde{\mu}\, g_2^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, \mathrm{LF}_{3,1,1,-1}\left[m_2, m_{\tilde{q}}^{p}, \tilde{\mu}\right] - \frac{3}{4} m_2\,\tilde{\mu}\, g_2^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, \mathrm{LF}_{2,2,1,-1}\left[m_2, \tilde{\mu}, m_{\tilde{q}}^{p}\right] - \\
&\frac{2}{3} m_3\,\tilde{\mu}\, g_3^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, \mathrm{LF}_{2,1,1,0}\left[m_3, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{i1}\right] - \frac{1}{3} m_3\,\tilde{\mu}\, g_3^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, \mathrm{LF}_{2,2,1,-1}\left[m_3, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{i1}\right] + \\
&\frac{2}{3} m_3\,\tilde{\mu}\, g_3^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, \mathrm{LF}_{3,1,1,-1}\left[m_3, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{i1}\right] + \frac{1}{3} m_3\,\tilde{\mu}\, g_3^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, \mathrm{LF}_{2,2,1,-1}\left[m_3, m_{\tilde{u}}^{i1}, m_{\tilde{q}}^{p}\right] - \\
&\frac{1}{8}\tilde{\mu}\, \overline{y_d}^{pr}\, \overline{y_u}^{si1}\, y_u^{pi2}\, a_d^{sr}\, \mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{d}}^{r}, \tilde{\mu}, m_{\tilde{q}}^{s}\right] + \\
&\frac{1}{8}\tilde{\mu}\, \overline{y_d}^{pr}\, \overline{y_u}^{si1}\, y_u^{pi2}\, a_d^{sr}\, \mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{q}}^{s}, \tilde{\mu}, m_{\tilde{d}}^{r}\right] + \frac{1}{3} m_1\,\tilde{\mu}\, g_1^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, \mathrm{LF}_{2,1,1,0}\left[\tilde{\mu}, m_1, m_{\tilde{u}}^{i1}\right] - \\
&\frac{1}{3} m_1\,\tilde{\mu}\, g_1^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, \mathrm{LF}_{3,1,1,-1}\left[\tilde{\mu}, m_1, m_{\tilde{u}}^{i1}\right] - \frac{1}{4}\tilde{\mu}\, \overline{y_d}^{pr}\, \overline{y_u}^{si1}\, y_u^{pi2}\, a_d^{sr}\, \mathrm{LF}_{2,1,1,0}\left[\tilde{\mu}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}\right] + \\
&\frac{1}{4}\tilde{\mu}\, \overline{y_d}^{pr}\, \overline{y_u}^{si1}\, y_u^{pi2}\, a_d^{sr}\, \mathrm{LF}_{3,1,1,-1}\left[\tilde{\mu}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}\right] - \frac{1}{6} m_1\,\tilde{\mu}\, g_1^2\, \overline{y_u}^{pi1}\, y_u^{pi2} \\
&\quad \mathrm{LF}_{1,1,1,1,0}\left[m_1, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{i1}, \tilde{\mu}\right] - \frac{1}{12} m_1\,\tilde{\mu}\, g_1^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, \mathrm{LF}_{2,1,1,1,-1}\left[m_1, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{i1}, \tilde{\mu}\right] - \\
&\frac{1}{6}\tilde{\mu}\, g_1^2\, \overline{y_u}^{pi1}\, a_u^{pi2}\, \mathrm{LF}_{1,1,1,1,0}\left[m_1, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{i2}, \tilde{\mu}\right] + \frac{1}{12}\tilde{\mu}\, g_1^2\, \overline{y_u}^{pi1}\, a_u^{pi2} \\
&\quad \mathrm{LF}_{2,1,1,1,-1}\left[m_1, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{i2}, \tilde{\mu}\right] - \frac{1}{4}\tilde{\mu}\, \overline{y_d}^{pr}\, \overline{y_u}^{si1}\, y_u^{pi2}\, a_d^{sr}\, \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, \tilde{\mu}\right] - \\
&\frac{2}{9}\tilde{\mu}\, g_1^2\, \overline{y_u}^{pi1}\, a_u^{pi2}\, \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{q}}^{p}, m_1, m_{\tilde{u}}^{i1}, m_{\tilde{u}}^{i2}\right] + \frac{1}{12} m_1\,\tilde{\mu}\, g_1^2\, \overline{y_u}^{pi1}\, y_u^{pi2} \\
&\quad \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{q}}^{p}, m_1, m_{\tilde{u}}^{i1}, \tilde{\mu}\right] - \frac{1}{12}\tilde{\mu}\, g_1^2\, \overline{y_u}^{pi1}\, a_u^{pi2}\, \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{q}}^{p}, m_1, m_{\tilde{u}}^{i2}, \tilde{\mu}\right] - \\
&\frac{2}{3}\tilde{\mu}\, g_3^2\, \overline{y_u}^{pi1}\, a_u^{pi2}\, \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{q}}^{p}, m_3, m_{\tilde{u}}^{i1}, m_{\tilde{u}}^{i2}\right] + \frac{1}{4}\tilde{\mu}\, \overline{y_u}^{pi1}\, \overline{y_u}^{rs}\, y_u^{ri2}\, a_u^{ps} \\
&\quad \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{u}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{r}, \tilde{\mu}\right] - \frac{1}{12} m_1\,\tilde{\mu}\, g_1^2\, \overline{y_u}^{pi1}\, y_u^{pi2}\, \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{u}}^{i1}, m_1, m_{\tilde{q}}^{p}, \tilde{\mu}\right] + \\
&\frac{1}{12}\tilde{\mu}\, g_1^2\, \overline{y_u}^{pi1}\, a_u^{pi2}\, \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{u}}^{i2}, m_1, m_{\tilde{q}}^{p}, \tilde{\mu}\right] + \frac{1}{12} m_1\,\tilde{\mu}\, g_1^2\, \overline{y_u}^{pi1}\, y_u^{pi2} \\
&\quad \mathrm{LF}_{2,1,1,1,-1}\left[\tilde{\mu}, m_1, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{i1}\right] - \frac{1}{12}\tilde{\mu}\, g_1^2\, \overline{y_u}^{pi1}\, a_u^{pi2}\, \mathrm{LF}_{2,1,1,1,-1}\left[\tilde{\mu}, m_1, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{i2}\right]\Big)
\end{aligned}
$$